

Question

Write a Java program to demonstrate single inheritance using extends.

The superclass A1 contains a variable and methods, and the subclass B1 inherits them. Calculate the sum of the inherited variable x and subclass variable y, and demonstrate access to superclass and subclass methods.

Steps for Execution

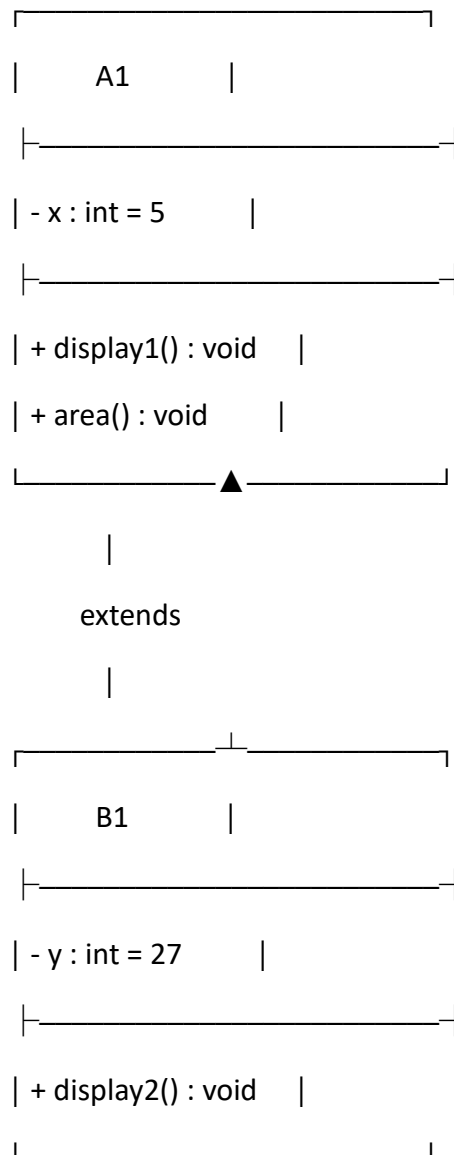
1. Create a class A1 as the superclass.
2. Declare integer variable x = 5 in A1.
3. Define display1() to display a superclass message.
4. Define area() to calculate and display the area of a square.
5. Create class B1 using extends A1.
6. Declare integer variable y = 27 in B1.
7. Define display2() to display a subclass message.
8. Create an object b1 of class B1 in the main() method.
9. Access the inherited variable x and subclass variable y.
10. Calculate x + y and display the result.
11. Call display1() and display2() using the b1 object.
12. Compile and execute the program.

Algorithm

1. **Start**
2. Define superclass A1 with x = 5.
3. Define methods display1() and area() in A1.
4. Define subclass B1 using extends A1.
5. Declare y = 27 and method display2() in B1.
6. In main(), create an object b1 of B1.
7. Compute result = b1.x + b1.y.
8. Display the value of result.
9. Invoke b1.display1().
10. Invoke b1.display2().

11. Stop

UML Class Diagram



Here, B1 is the **subclass** and A1 is the **superclass**. The arrow represents inheritance.

Code

```
package javapractice;
```

```
class A1 {
```

```
    int x = 5;
```

```
    void display1() {
```

```
        System.out.println("this is superclass");  
    }
```

```
    void area() {  
        int side = 3;  
        System.out.println("area of square is:" + (side * side));  
    }  
}
```

```
class B1 extends A1 {  
    int y = 27;  
  
    void display2() {  
        System.out.println("this is subclass");  
    }  
}
```

```
public class C1ExtendsC2 {  
    public static void main(String[] args) {  
        B1 b1 = new B1();  
  
        int result = b1.x + b1.y;  
        System.out.println("result is:" + result);  
  
        b1.display1();  
        b1.display2();  
    }  
}
```

Output

result is:32

this is superclass

this is subclass